

Interim Report

Machine Learning Prediction of Restaurant Ingredient Demand for Stock Management and Food Wastage Control

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1. Introduction

In order to enhance supply management and minimise food waste, the study explores whether machine learning algorithms can reliably forecast restaurant ingredient demand. Inaccurate demand estimation may make it difficult for the food service business to balance ingredient availability while minimising waste. While understocking may result in lost sales and lower customer satisfaction, overstocking may cause wastage.

The project aims to create a predictive system that uses past restaurant data to forecast demand at the ingredient level. It aims to enhance efficiency, assist data-driven decision-making in restaurant operations and advance sustainability goals by showing these forecasts as useful insights via a dashboard.

2. Background/ Literature Review

Being part of the food service industry sector, working as a server, one of the challenges faced most by restaurants is food management. With the help of this research, it would be able to show whether with the use of predictive models, overcomes this challenge as it would work upon the historical data and predict how can the stock be managed for future. It would help in predicting the next popular items and the items that need restocking and stored for the upcoming days for small – size restaurants.

Traditional management systems are often based on the past average sales or human estimation, which sometimes fail to consider varied factors. With the help of artificial intelligence and data science, the use of predictive analysis for demand forecasting has been possible.

Recent research has explored data- driven solutions to the issue through demand forecasting. For instance, there has been research done for predicting the number of customers with the help of models like Bayesian Linear Regression, Boosted Decision Tree, and Decision Forest Regression (Tanizaki, T., Hoshino, T., Shimmura, T. and Takenaka, T., 2019). Another research has been implemented for improving sales forecast accuracy (Sakib, S.N., 2023) and research for demand forecasting model for predicting the

daily demand of fresh food products in food retail market (Seyam, A. et al. (2025)).

A research based on encouraging proper employee scheduling for managing crew load, restaurants need accurate sales forecasting (Schmidt, A., Kabir, M.W.U. and Hoque, M.T., 2022) developed the models based on LSTM and TFT giving an overall best performance with errors less than 20%.

This research builds upon developing machine learning based model that predicts ingredient demand, visualizes the outcome through a dashboard and the potential to reduce food waste and improve operational efficiency.

3. Aim and Objectives

Aim

To develop and evaluate a machine learning based predictive model that forecasts restaurant ingredient demand to optimise stock management decisions and reduce food waste.

Objectives

1. To collect and prepare restaurant order and ingredient usage data suitable for time-series and machine learning based forecasting.
2. To design, train and compare predictive models using Python, measuring performance with RMSE, MAE and R2.
3. To analyse key influencing factors such as weather using feature importance techniques.
4. To develop a dashboard that visualises ingredient demand forecasts and helps with stock management decisions.
5. To evaluate the potential reduction in forecast error, ingredient overstock and food waste achieved through the proposed solution.

4. Methodology

This research uses an experimental, data-driven methodology based on time-series analysis and machine learning. It makes use of historical restaurant transaction data which includes menu items, ingredients, quantities sold, weather and promotional factors.

Cleaning of data, changing it into the forms of weekly demand and feature engineering are examples of how data preprocessing can be done.

Exploratory data analysis (EDA) is carried out to find patterns, seasonality and different demand drivers. Python libraries like pandas, scikit-learn, Prophet and XGBoost are used to train the forecasting models.

So as to include external factors for the same, machine learning model such as XGBoost has been utilised after a baseline time-series model, Prophet, to capture trends and seasonality. Feature importance is also used to address model interpretability.

5. Implementation

Google colab is used for the Python implementation, while Visual Studio Code facilitates development. The project is executed locally on a standard laptop.

The following implementation steps have been completed as of now-

- Development of ingredient-level datasets using menu-level data.
- Weekly total of the demand for ingredients
- Preprocessing and data cleaning
- Investigative examination of patterns in ingredient demand
- Prophet model implementation for multiple substances
- Analysing baseline model prediction with RMSE and MAE

Power BI/Tableau dashboards that show predicted demand, confidence intervals and inventory suggestions are coupled to forecast outputs that are exported as CSV files. This can help in showing that prediction results are converted into useful insights for stock management.

6. Testing / Evaluation Strategy

The main goal of evaluation is to determine how successfully the prediction models accomplish the projects goals. In order to maintain timeline and predict forecasting conditions, train-test splits are used. RMSE, MAE and R” are used to measure the performance of the model.

Forecast error reduction is measured by comparing the baseline (Prophet) with machine learning models (XGBoost). When forecasts are utilised for inventory planning, business-oriented evaluation is carried out by predicting possible decreases in overstocking and food waste.

By graphically contrasting actual and projected demand and emphasizing dashboard outputs enhance assessment.

7. Professional , Social , Economic and Legal Issues

Reducing food waste promotes sustainable resource use and has major positive effects on the environment and society. By cutting down on wasteful expenses, better stock management can help increase restaurant profitability. Data privacy (if actual restaurant data is used) and compliance with GDPR for any personal or transactional data are the legal and ethical considerations.

The project will align with the codes of ethics, emphasising on fairness, transparency, and responsible AI techniques. Anonymous or synthetic data will be used to guarantee adherence to the moral principles.

8. Resource Requirements

To create the proposed machine learning models and dashboards, the following tools and accessories can be used and accessed-

- Hardware: Standard laptop and university labs for model training and analysis.
- Software: Python (pandas, scikit-learn, numpy, prophet, XGBoost, matplotlib, flask), Power BI/ Tableau
- Datasets: Publicly available sources such as Kaggle or similar restaurant sales datasets

No specialised hardware or cloud infrastructure is required.

9. References/Information Sources

Seyam, A. et al. (2025) A stacking ensemble model for food demand forecasting: A preventative approach to food waste reduction. *Cleaner Logistics and Supply Chain*. [Online] fifteen.

Tanizaki, T., Hoshino, T., Shimmura, T. and Takenaka, T., 2019. Demand forecasting in restaurants using machine learning and statistical analysis. *Procedia CIRP*, 79, pp.679-683.

Sakib, S.N., 2023. Restaurant sales prediction using machine learning. In *Handbook of research on AI and machine learning applications in customer support and analytics* (pp. 202-226). IGI Global.

Schmidt, A., Kabir, M.W.U. and Hoque, M.T., 2022. Machine learning based restaurant sales forecasting. *Machine Learning and Knowledge Extraction*, 4(1), pp.105-130.

Anon (2022) *Restaurant Sales Prediction Using Machine Learning*. NewsRX LLC.